```python
import pandas as pd

# ---------------- CREATE / READ -----------------
# Load dataset
df = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Amit', 'Riya', 'John'],
    'Marks': [85, 90, 78]
})

print("Original Dataset:")
print(df)

# ---------------- CREATE -----------------
new_row = {'ID': 4, 'Name': 'Sneha', 'Marks': 88}
df = pd.concat([df, pd.DataFrame([new_row])], ignore_index=True)

print("\nAfter CREATE (New Row Added):")
print(df)

# ---------------- READ -----------------
print("\nREAD Operation - Displaying first 3 rows:")
print(df.head(3))

# ---------------- UPDATE -----------------
df.loc[df['ID'] == 2, 'Marks'] = 95

print("\nAfter UPDATE (Marks of ID 2 updated):")
print(df)

# ---------------- DELETE -----------------
df = df[df['ID'] != 3]

print("\nAfter DELETE (Record with ID 3 removed):")
print(df)
```

```
Original Dataset:
   ID  Name  Marks
0   1  Amit     85
1   2  Riya     90
2   3  John     78

After CREATE (New Row Added):
   ID   Name  Marks
0   1   Amit     85
1   2   Riya     90
2   3   John     78
3   4  Sneha     88

READ Operation - Displaying first 3 rows:
   ID  Name  Marks
0   1  Amit     85
1   2  Riya     90
2   3  John     78

After UPDATE (Marks of ID 2 updated):
   ID   Name  Marks
0   1   Amit     85
1   2   Riya     95
2   3   John     78
3   4  Sneha     88

After DELETE (Record with ID 3 removed):
   ID   Name  Marks
0   1   Amit     85
1   2   Riya     95
3   4  Sneha     88
```

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Sample dataset for classification
data = {
    'Age': [22, 25, 47, 52, 46, 56, 55, 60],
    'Salary': [15000, 22000, 35000, 42000, 38000, 52000, 51000, 61000],
    'Bought_Insurance': [0, 0, 1, 1, 1, 1, 1, 1]
}

df = pd.DataFrame(data)
```

```
# Split data
X = df[['Age', 'Salary']]
y = df['Bought_Insurance']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)

# Logistic Regression model
model = LogisticRegression()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Predicted Values:", y_pred)
print("Actual Values:", list(y_test))
print("Accuracy:", accuracy)
```

```
Predicted Values: [0 1]
Actual Values: [0, 1]
Accuracy: 1.0
```

```
Start coding or generate with AI.
```